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EXPERIMENT 04

Data modelling and form design

Aim

Design a normalized library data model and implement forms with appropriate types, keys, relationships, and validation.

Procedure

1. Create the four core forms below. Use the exact link names if using the supplied Deluge. Creator supplies a record ID; do not replace it with a user-entered code.

2. Use a single-select lookup for each relationship: Book_Copies.Book to Books, Loans.Copy to Book_Copies, and Loans.Member to Members. Thus Books 1:N Book_Copies, Book_Copies 1:N Loans over time, and Members 1:N Loans.

3. Set mandatory fields, allowed choices and duplicate restrictions as specified. Restrict editing of identifiers after related records exist. Do not delete titles, copies or members with loan history; mark them inactive or out of service.

Form	Fields and types
Books	Book_Code: unique required text; Title: required text; Author: text; ISBN: optional text; Category: dropdown. ISBN stays text to preserve leading zeros.
Book_Copies	Accession_No: unique required text; Book: required lookup to Books; Shelf: text; Service_Status: required dropdown, In Service / Maintenance / Lost.
Members	Member_Code: unique required text; Full_Name: required single-line text; Email: required email; Active: decision box; Portal_Login: optional email; CRM_Contact_ID: text; CRM_Sync_Status: dropdown Pending / Synced / Error.
Loans	Loan_No: auto number; Copy: required lookup; Member: required lookup; Issue_Date: required date; Due_Date: required date; Return_Date: optional date; Status: required dropdown Issued / Returned; Fine_Amount: currency, default 0; Last_Reminder_On: optional date.

Identity and derived data

Keep one Members record per person. For this lab, require unique Email and, where used, a one-to-one Portal_Login mapping. Email is not a universal real-world identity key; document shared-address and address-change handling before deployment.

A copy is available when Service_Status is In Service and no Loans record for that Copy has Status Issued. An overdue loan has Status Issued and Due_Date earlier than the reference date. Do not store an independently editable Available_Count or Overdue flag that can drift from the loan history.

Implementation and verification

Procedure continued

1. Arrange form sections as identification, descriptive data and administrative data. Display copy accession number with the book title in the issue lookup. Use clear labels and help text; keep link names stable.

2. In the Loans create workflow, set Issue_Date to today, Due_Date to today plus 14 days, Status to Issued and Fine_Amount to zero. Enforce these values on the server; disabling a field in the UI is only a usability measure.

3. Permit new loans only for active members and in-service copies. Reject a fourth open loan and any copy with an existing open loan. Put these checks in On Validate as detailed in Experiment 6.

4. Restrict general editing of Loans. Route returns through a librarian-only return action or a controlled edit workflow that permits only the return transition. Historical dates, member and copy must not be casually changed.

5. Create the synthetic fixtures below. Use additional copies when testing the three-loan limit. Never import real member personal data merely to populate the lab.

Entity	Synthetic records
Books	B001 Database Systems; B002 Introduction to Programming; B003 Software Engineering.
Copies	C001 and C002 → B001; C003 → B002; C004 → B003. Set all In Service initially, then use C004 for maintenance tests.
Members	M001 Asha Test, asha@example.com, Active true; M002 Ravi Test, ravi@example.com, Active true; M003 Inactive Test, inactive@example.com, Active false.
Loan fixture	Issue C001 to M001 on D. Expected due date D + 14. For overdue tests, create a separate authorized historical fixture in development only.

Testing and evidence

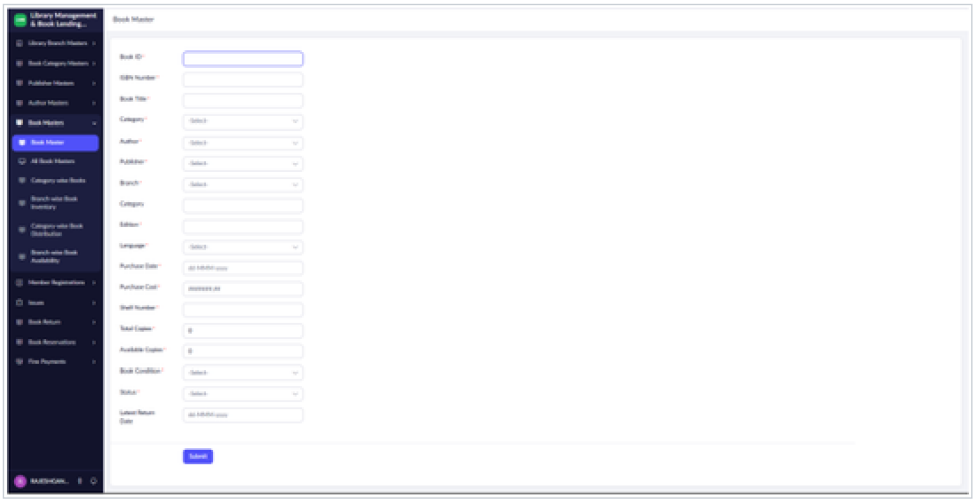
E04: relationship diagram, field dictionary, form screenshots and test records. Try a duplicate accession number, missing book lookup, invalid email and inactive member. Expect rejection. Change a title and confirm its linked copies still reference the same Books ID. Attempt to delete a referenced member and verify the chosen restriction.

Record actual Creator record IDs alongside fixture labels; C001 is a display code, not a numeric lookup ID. Check one book with two copies and two sequential loans of the same returned copy.

EXPERIMENT 04 VISUAL RECORD

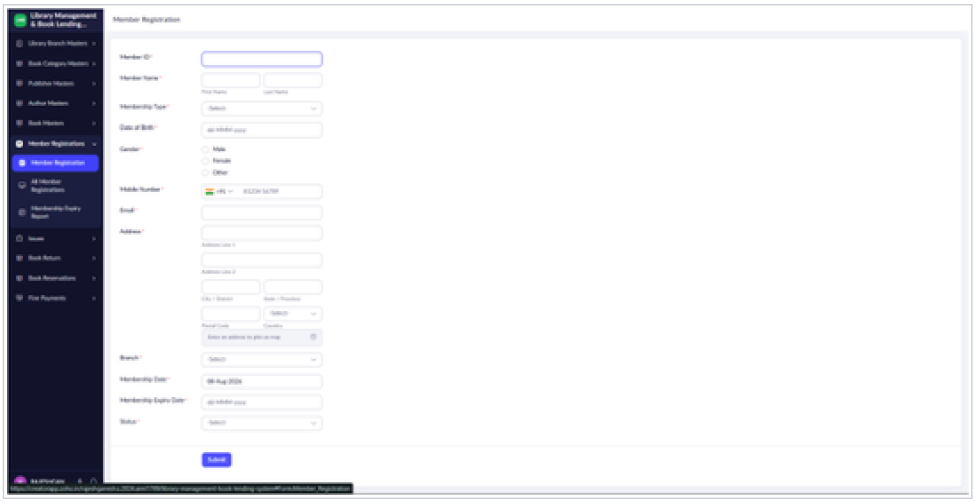
Application images and explanation

Book Master entry form



Supplied application screenshot from the uploaded library document.

Member Registration entry form



Supplied application screenshot from the uploaded library document.

The supplied app uses named master forms and separate Book Issue and Book Return screens. The written examples use a normalized Books, Book_Copies, Members and Loans model. Treat these as illustrative link names and map the actual fields before using scripts. Preserve the link between an issue and its return.

Result

Expected result and verification criteria. All four forms preserve the intended relationships and input constraints. Fixtures can be traced across lookups, invalid input is rejected, and existing loan history is protected. No duplicate-copy allocation guarantee is claimed solely from a lookup filter.